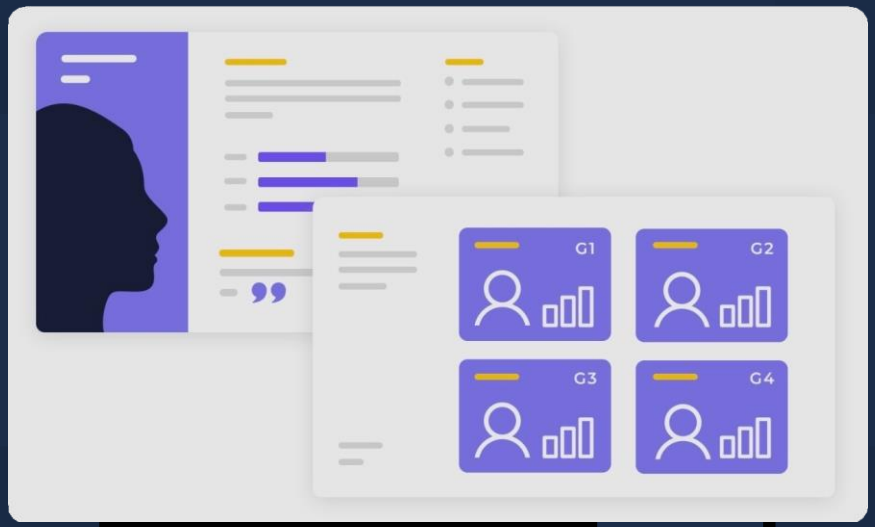

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TRAVELFLOW

JOURNEY BEYOND THE HORIZON

USER SCENARIOS

This document presents the main user scenarios for the TravelFlow system, describing how different users interact with the platform to plan, manage, and enhance their travel experience.

Client Scenarios

TRAVELFLOW — USER SCENARIOS

SCENARIO 1

User Registration

SCENARIO: *Ana, a new visitor, creates a TravelFlow client account from scratch.*
ACTORS: New Visitor (Ana), System

PRE-CONDITIONS

- The TravelFlow website is accessible and operational.
- Ana has a valid email address not yet registered in the system.
- No Users record exists with Ana's email.

MAIN FLOW

1. Ana opens the TravelFlow homepage in her browser.
2. She clicks 'Login' in the navigation bar; the auth modal appears.
3. She selects the 'Register' tab and fills in full name, email, and password.
4. She clicks 'Create Account'.
5. The system validates all fields for completeness and email format.
6. A new Users record is inserted with role = 'client'.
7. A welcome toast confirms successful registration; Ana is logged in automatically.

ALTERNATIVE FLOW

- If a required field is empty, it is highlighted and submission is blocked.
- If the email format is invalid, an inline error message appears.
- If the email already exists, Ana is shown an appropriate error.
- If the password is too short (< 6 chars), a strength warning is displayed.

POST-CONDITIONS

- A Users record is saved with role = 'client'.
- Ana is authenticated and all client-level features are accessible.
- The navigation bar reflects the logged-in state.

SCENARIO 2

User Login

SCENARIO: *Registered user Marco logs into his existing TravelFlow account.*
ACTORS: Registered User (Marco), System

PRE-CONDITIONS

- Marco has a valid Users record in the database.
- Marco is currently logged out.

MAIN FLOW

1. Marco clicks 'Login' in the navigation bar.
2. He enters his registered email and password.
3. The system retrieves the matching Users record and compares credentials.
4. Authentication succeeds; a session is created and stored client-side.
5. Marco's user badge appears in the navigation bar.
6. He is returned to the page he was viewing before login.

ALTERNATIVE FLOW

- If the email does not exist, a generic 'Invalid email or password' error is shown.
- If the password is wrong, the same generic error appears (no field hint, for security).

POST-CONDITIONS

- Marco's session is active and tied to his Users record.
- The navigation bar reflects the authenticated state.
- Marco can access all client-level pages and features.

SCENARIO 3

User Logout

SCENARIO: *Authenticated client Sara ends her session safely.*
ACTORS: Authenticated Client (Sara), System

PRE-CONDITIONS

- Sara is currently logged in with an active session.
- Sara has finished her tasks on the platform.

MAIN FLOW

1. Sara clicks her user badge in the top-right corner of the navigation bar.
2. A dropdown menu appears with options: My Account and Log Out.
3. Sara clicks 'Log Out'.
4. The system clears the session data.
5. The navigation bar reverts to showing Login / Book Now buttons.

ALTERNATIVE FLOW

- If Sara closes the browser without logging out, the session expires after timeout.
- If Sara navigates to a protected page after logout, she is redirected to login.

POST-CONDITIONS

- Sara's session is cleared from client storage.
- All client-only UI elements are hidden.
- Sensitive booking data is no longer accessible without re-authentication.

SCENARIO 4

Searching for Available Trips

SCENARIO: *Client David searches for buses from Tirana to Himara on a specific date.*
ACTORS: Authenticated Client (David), System

PRE-CONDITIONS

- David is logged in.
- At least one trip exists for the requested route and date with status = 'scheduled'.

MAIN FLOW

1. David navigates to the 'Book' page via the navigation bar.
2. The search panel displays three fields: From, To, and Date.
3. David selects 'Tirana' → 'Himara' and picks a travel date.
4. The system maps the city pair to the route_id and queries the Trips table.
5. If no trips exist for that date, autoTrips generates them automatically.
6. Matching trips render as cards showing route, departure/arrival time, bus model, and price.
7. David clicks on his preferred trip to proceed to booking.

ALTERNATIVE FLOW

- If no trips match, a 'No trips found – try a different date' message appears.
- If From and To cities are identical, a validation error prevents the query.
- David can reset the form and enter new criteria at any time.

POST-CONDITIONS

- David sees a filtered list of available trips.
- No records are created or modified in the database.
- David is ready to proceed to the booking flow.

SCENARIO 5

Booking a Ticket with Seat Selection

SCENARIO: *Client Elena books a ticket, manually picking her preferred seat and destination stop.*
ACTORS: Authenticated Client (Elena), System

PRE-CONDITIONS

- Elena is logged in.
- Elena has completed a trip search and sees a results list.
- At least one available seat exists on the selected trip.

MAIN FLOW

1. Elena clicks 'Book' on a trip card.
2. The booking page loads with full trip details.
3. Elena selects her destination stop; the price updates incrementally.
4. She chooses 'Pick My Seat'; an interactive seat map appears (red = occupied, green = free).
5. She clicks a seat; it turns teal to confirm selection.
6. She clicks 'Show Fee Breakdown' to verify itemised costs.
7. She clicks 'Confirm Booking'.
8. The system validates availability, inserts a Bookings record with a unique code, and updates bookedSeats.
9. A toast notification confirms the booking with the code.

ALTERNATIVE FLOW

- If Elena skips stop selection, 'Confirm Booking' remains disabled.
- If the seat is taken concurrently, Elena is prompted to choose again.
- If the trip fills up between search and booking, a 'Sold Out' notice appears.

POST-CONDITIONS

- A Bookings record exists with status = 'confirmed' and a unique booking code.
- The seat is locked in bookedSeats, preventing double-booking.
- Elena can view her ticket under 'My Account'.

SCENARIO 6

Booking a Ticket with Random Seat

SCENARIO: *Client Joni books a ticket quickly without caring which specific seat he gets.*
ACTORS: Authenticated Client (Joni), System

PRE-CONDITIONS

- Joni is logged in.
- A trip with available seats exists.

MAIN FLOW

1. Joni searches for and selects a trip.
2. He selects his destination stop.
3. He selects 'Random Seat' from the seat options.
4. A message confirms: 'A seat will be assigned automatically at no extra charge.'
5. The price summary shows ticket price + site fee (100 lek) only.
6. Joni clicks 'Confirm Booking'.
7. The system assigns the lowest available seat number and inserts the Bookings record.

ALTERNATIVE FLOW

- If all seats are occupied, booking is blocked and an error is shown.
- Joni can switch to 'Pick My Seat' mode at any point.

POST-CONDITIONS

- A confirmed Bookings record is created with a system-assigned seat number.
- Joni's ticket displays the assigned seat number.

SCENARIO 7

Viewing Booking History and Ticket

SCENARIO: *Client Luca reviews his past and upcoming bookings and opens a digital ticket.*
ACTORS: Authenticated Client (Luca), System

PRE-CONDITIONS

- Luca is logged in.
- Luca has at least one booking record in the Bookings table.

MAIN FLOW

1. Luca clicks his user badge and selects 'My Account'.
2. The page loads a chronological list of all his bookings.
3. Each card shows: booking code, route, trip date, seat, stop, total price, and status.
4. Luca clicks 'Show Ticket' on a confirmed booking.
5. The digital ticket is displayed with all travel details.

ALTERNATIVE FLOW

- If Luca has no bookings, an empty-state message is shown with a link to search.
- Luca can filter bookings by status: All, Confirmed, Cancelled.
- Tickets for past trips remain viewable for reference.

POST-CONDITIONS

- No database records are modified.
- Luca has complete visibility of his travel history.

SCENARIO 8

Cancelling a Booking

SCENARIO: *Client Mira can no longer travel and wants to cancel her upcoming reservation.*
ACTORS: Authenticated Client (Mira), System

PRE-CONDITIONS

- Mira is logged in.
- Mira has a booking with status = 'confirmed'.
- The trip has not yet departed.

MAIN FLOW

1. Mira navigates to 'My Account'.
2. She locates the booking she wants to cancel and clicks 'Cancel'.
3. A confirmation dialog appears: 'Are you sure? This action cannot be undone.'
4. Mira confirms; the system updates the Bookings record status to 'cancelled'.
5. The seat is removed from bookedSeats in the Trips table.
6. A toast notification confirms the cancellation.

ALTERNATIVE FLOW

- If Mira clicks 'Go Back' in the dialog, no changes are saved.
- If the booking is already 'cancelled', the cancel button is not shown.
- If the departure time has passed, the cancel button is disabled.

POST-CONDITIONS

- The Bookings record status is updated to 'cancelled'.
- The seat is released in bookedSeats and available for other users.
- Mira's account page reflects the updated status.

SCENARIO 9

Renting a Car

SCENARIO: *Client Sofia browses the rental fleet and confirms a car rental reservation.*
ACTORS: Authenticated Client (Sofia), System

PRE-CONDITIONS

- Sofia is logged in.
- At least one car with available = 1 exists in the Cars table.

MAIN FLOW

1. Sofia clicks 'Rentals' in the navigation bar.
2. The catalog displays available cars as cards with make, model, year, category, fuel, transmission, seats, and daily price.
3. Sofia applies filters to narrow results.
4. She clicks 'Rent'; a form appears with pickup date, return date, and pickup location.
5. The system calculates the number of rental days and total price.
6. Sofia confirms; a Rentals record is inserted with status = 'confirmed' and the car's available flag is set to 0.

ALTERNATIVE FLOW

- If no cars match the filters, a 'No vehicles available' message is shown.
- If the return date is on or before the pickup date, a validation error appears.
- If a car becomes unavailable between browsing and confirming, an error appears.

POST-CONDITIONS

- A Rentals record is created with status = 'confirmed'.
- The rented car's available flag is set to 0 in the Cars table.
- Sofia can view the rental in 'My Account'.

SCENARIO 10

Viewing Rental History

SCENARIO: *Client Arben reviews his car rental reservations in his account page.*
ACTORS: Authenticated Client (Arben), System

PRE-CONDITIONS

- Arben is logged in.
- Arben has at least one entry in the Rentals table.

MAIN FLOW

1. Arben navigates to 'My Account'.
2. He scrolls to the 'My Rentals' section.
3. A list of all his rentals is shown: make/model, pickup date, return date, total price, and status.
4. Arben clicks on a rental card to view full details.

ALTERNATIVE FLOW

- If Arben has no rentals, the section shows 'No rentals yet – browse our fleet.'
- Arben can filter by status: Confirmed, Completed.

POST-CONDITIONS

- No database records are modified.
- Arben has clear visibility of all his rental reservations.

Driver Scenarios

TRAVELFLOW — USER SCENARIOS

SCENARIO 11

Driver Accessing the Driver Portal

SCENARIO: *Bus driver Petrit logs in and opens the driver-specific portal.*
ACTORS: Bus Driver (Petrit), System

PRE-CONDITIONS

- Petrit has a Users record with role = 'driver' and an assigned bus_id and route.
- Petrit is currently logged out.

MAIN FLOW

1. Petrit opens TravelFlow and clicks 'Login'.
2. He enters his driver email and password.
3. The system authenticates him and detects role = 'driver'.
4. The navigation bar shows the 'Driver' link (hidden for clients and admins).
5. The Driver Portal loads showing his assigned trips with departure time, bus plate, and passenger count.

ALTERNATIVE FLOW

- If Petrit accesses the Driver page without a valid session, he is redirected to login.
- If a client manually navigates to the driver URL, 'Access Denied' is shown.
- If no trips are assigned, a 'No trips assigned – contact your administrator' message appears.

POST-CONDITIONS

- Petrit has access to all driver tools and his trip schedule.
- Admin pages remain restricted for drivers.

SCENARIO 12

Driver Validating a Passenger Ticket

SCENARIO: *Driver Petrit validates passenger booking codes before departure.*
ACTORS: Bus Driver (Petrit), System

PRE-CONDITIONS

- Petrit is logged in with role = 'driver'.
- At least one passenger holds a confirmed booking for this trip.

MAIN FLOW

1. Petrit opens the Driver Portal and selects the ticket validation panel.
2. A booking code input field appears.
3. The passenger presents their booking code.
4. Petrit enters the code and submits.
5. The system queries the Bookings table by the uppercase booking code.
6. If found and confirmed, a green panel shows: passenger name, seat number, stop name, and route.
7. The booking status is updated to 'used'.
8. Petrit can validate multiple passengers consecutively without refreshing.

ALTERNATIVE FLOW

- If the code is not found, a red 'NOT FOUND' error panel is shown.
- If the ticket was already validated, 'ALREADY VALIDATED' warns Petrit of a potential duplicate.
- If the booking belongs to a different trip, a mismatch error is displayed.

POST-CONDITIONS

- The booking status is updated to 'used' in the Bookings table.

Admin Scenarios

TRAVELFLOW — USER SCENARIOS

SCENARIO 13

Admin Accessing the Dashboard

SCENARIO: *Administrator Drita logs in and accesses the full admin management dashboard.*
ACTORS: Admin (Drita), System

PRE-CONDITIONS

- Drita has a Users record with role = 'admin'.
- Drita is currently logged out.

MAIN FLOW

1. Drita logs in with admin credentials.
2. The system detects role = 'admin'; the 'Admin' link appears in the navigation bar.
3. The Admin Dashboard loads, showing panels for: Buses, Cars, Users, Rentals, Maintenance, and Statistics.
4. The overview shows: confirmed bookings, total revenue, total users, total buses, and active rentals.

ALTERNATIVE FLOW

- A client navigating manually to the admin URL is shown 'Access Denied'.
- If a panel fails to load, a 500 error page with a retry option is displayed.

POST-CONDITIONS

- Drita has full access to all administrative functions.
- Non-admin users remain unable to access any admin panel.

SCENARIO 14

Admin Managing Buses

SCENARIO: *Admin Mark views the bus fleet and registers a new bus.*
ACTORS: Admin (Mark), System

PRE-CONDITIONS

- Mark is logged in as admin.

MAIN FLOW

1. Mark opens the Admin 'Buses' panel; a table lists all buses with plate, model, capacity, route, and status.
2. Mark clicks 'Add Bus'.
3. A form appears: Bus Code, Plate Number, Model, Capacity, Route, and Status.
4. Mark fills in all fields and submits.
5. The system validates uniqueness of bus code and plate and inserts a new Buses record.
6. A success notification confirms; the new bus is available for its assigned route.

ALTERNATIVE FLOW

- If the bus code or plate already exists, a uniqueness error is shown.
- Mark can set the initial status to 'inactive' if the bus is not yet road-ready.

POST-CONDITIONS

- A new Buses record exists in the database.
- The bus is visible in all admin fleet views.

SCENARIO 15

Admin Managing the Rental Car Fleet

SCENARIO: *Admin Greta registers a new car so clients can rent it.*
ACTORS: Admin (Greta), System

PRE-CONDITIONS

- Greta is logged in as admin.
- The new car's code and plate are not already registered.

MAIN FLOW

1. Greta opens the Admin 'Cars' panel; all cars are listed with details and availability.
2. She clicks 'Add Car'.
3. A form appears: Car Code, Make, Model, Year, Plate, Price Per Day, Category, Fuel, Transmission, Seats, and optional Photo URL.
4. Greta fills in all fields and submits.
5. The system validates uniqueness and inserts a Cars record with available = 1.
6. A success notification confirms; the car appears in the client-facing rental catalog.

ALTERNATIVE FLOW

- If the car code or plate already exists, a uniqueness error is displayed.
- Greta can set available = 0 if the car is temporarily unavailable.

POST-CONDITIONS

- A Cars record is created with available = 1.
- The car is immediately visible in the client-facing rental catalog.

SCENARIO 16

Admin Managing Rental Returns

SCENARIO: *Admin Greta processes the return of a rented car and makes it available again.*
ACTORS: Admin (Greta), System

PRE-CONDITIONS

- Greta is logged in as admin.
- A Rentals record with status = 'confirmed' exists and the return date has been reached.

MAIN FLOW

1. Greta opens the Admin 'Rentals' panel.
2. Active rentals are listed with car details, client name, pickup date, and return date.
3. Greta locates the rental and clicks 'Mark as Returned'.
4. The system updates the Rentals record status to 'completed'.
5. The car's available flag is reset to 1.
6. The car reappears in the client-facing rental catalog.

ALTERNATIVE FLOW

- If the return is delayed, Greta can extend the return date.
- The system's automatic checkRentals process also marks overdue confirmed rentals as completed daily.

POST-CONDITIONS

- The Rentals record status is 'completed'.
- The car is available again for new rental reservations.

SCENARIO 17

Admin Managing Maintenance Records

SCENARIO: *Admin Mark reviews the maintenance schedule and marks a service as completed for a bus.*
ACTORS: Admin (Mark), System

PRE-CONDITIONS

- Mark is logged in as admin.
- At least one maintenance record exists in the Maintenance table.

MAIN FLOW

1. Mark opens the Admin 'Maintenance' panel.
2. All vehicles are listed with their service schedules, last service date, and next due date.
3. Mark identifies an overdue service and checks its checkbox.
4. The system updates lastDate to today and calculates nextDate based on intervalDays.
5. Mark can also click 'Mark All' to mark all services for a vehicle as completed at once.
6. The updated services list is returned and reflected immediately in the panel.

ALTERNATIVE FLOW

- If the bus is assigned to an upcoming trip, the system warns Mark before changing its status.
- Multiple service types can be updated independently (oil, tyres, brakes, inspection, insurance, registration).

POST-CONDITIONS

- The Maintenance record is updated with the new lastDate and nextDate for each checked service.
- The next due date is automatically calculated and visible on the panel.

SCENARIO 18

Admin Creating a Driver Account

SCENARIO: *Admin Elena creates a new driver user account and links it to a bus.*
ACTORS: Admin (Elena), System

PRE-CONDITIONS

- Elena is logged in as admin.
- The new driver's email is not already registered in the Users table.

MAIN FLOW

1. Elena opens the Admin 'Users' panel and clicks 'Add Driver'.
2. A form requests: full name, email, initial password, assigned bus_id, and route.
3. Elena fills in all fields and submits.
4. The system validates email uniqueness and inserts a Users record with role = 'driver', bus_id, and route set.
5. The new driver can immediately log in and access the Driver Portal.

ALTERNATIVE FLOW

- If the email already exists, a uniqueness error is shown.
- Elena can deactivate a driver account without deleting it, preserving historical records.

POST-CONDITIONS

- A Users record with role = 'driver' is created and linked to a bus.
- The driver can log in and access all driver tools.

SCENARIO 19

Admin Viewing System Statistics

SCENARIO: *Admin Drita reviews platform-wide statistics to monitor performance.*
ACTORS: Admin (Drita), System

PRE-CONDITIONS

- Drita is logged in as admin.
- The system has processed bookings, rentals, and registered users.

MAIN FLOW

1. Drita opens the Admin Dashboard.
2. The statistics panel shows: total registered users, total confirmed bookings, total revenue (in lek), active rentals, and total buses.
3. Drita reviews the counters to get a quick performance snapshot.

ALTERNATIVE FLOW

- If no bookings or rentals exist, all counters display 0.

POST-CONDITIONS

- Drita has a clear picture of platform performance.
- No database records are modified; this is a read-only view.